ASEN 3801 Lab 3: Attitude Sensors, Actuators, And Spacecraft Pitch Controls

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INTRODUCTION

This lab introduces us to the key attitude sensors and actuators (The Rate Gyro, Control Moment Gyro and a Reaction Wheel) and how their combinations are built into the MEMS circuit that drives the spin module. The lab focuses on using the rate gyro, spin module hardware components along with bicycle wheels independently to have better interpretation of moment, torque, created due to rotating wheel, angular momentum, angular velocity and tracking the behavior of spin along with variable controls fed into using LABVIEW VI software. In reality, the change in angular momentum helps define the reorientation of the spacecraft after each successful CMG spin while we are interpreting this with the small actuators and sensors in the lab. The use of CMG lets us analyze the force that acts along a rotating wheel perpendicular to the radius of the wheel creating a certain torquer moment and interpreting that with our body's orientation to help better analyze the forces that our outstretched arms must act against to resist the torque created by the rotating wheel. The use of LABVIEW VI will allow us to get the data from the manual (w/ external force) and automatic (w/o external force) driven spin module and observe the sinusoidal waveform into LABVIEW VI to have a better understanding of the spin. The dataset recorded is analyzed further using MATLAB and to be compared with the ("true") data.

PROBLEM 1: PHYSICAL RATE GYRO

a) Rotating the Gyro Box

When the gyro box is rotated about the x-axis (roll), little effect was noticed or felt. The internal gyro rotation seemed to be from the external torque we were applying on the system by rotating it.

When the gyro box is rotated about the y-axis (pitch), no difference was noticed or felt.

When the gyro box was rotated about the z-axis (yaw), several effects were observed. Firstly, a noticeable resistance to a change in yaw was felt as we rotated about the z-axis. Secondly, the gyro would rotate in the x-axis, and in the opposite direction (a positive yaw resulted in a negative roll and vice versa).

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b) Determine Rotational Direction

When the box was rotated about the positive z-axis, the gyro tilted about the negative x-axis, and when rotated about the negative z-axis, the gyro tilted about the positive x-axis. This behavior demonstrates gyroscopic precession, where the response occurs perpendicular to the applied rotation. According to Euler's rotational equation, $\overline{M} = \overline{\Omega}\ x\ \overline{H}$. The applied angular velocity about the z-axis ($\overline{\Omega} =+\ z$) crossed with the rotors angular momentum vector ($\overline{H}$) determines the rotation of the torque. Since the observed torque was about -x for a + z rotation, the angular momentum must be along the + y-axis. This means that the rotor is spinning in a clockwise direction when viewed from the + y-axis

c) What Happens When the Rate Gyro Rotes About the Z - Axis?

The gyro wheel rotation about the y-axis would be tilting the gyroscope and where the resistive forces/moment about the same axis of rotation was felt at the inclination resisting further. The gyro wheel rotation about the x-axis would be rolling the gyroscope to observe the resistive force/moment resisting further motion in the same direction of rotation and we felt that the gyroscope tilted to align with the roll but there was minimal force/moment resisting the further motion in the -x or +x direction. The movement about z-direction for the gyroscope was most interesting as it didn't have any resistive motion because the circular motion of the gyroscope was parallel or anti-parallel to the yaw moment that corresponds to the +z or -z direction.

$$[I]\dot{\omega} = -[\omega][I]\omega + L_c$$

$$I_{xx}\dot{\omega_x} = -\left(I_{zz} - I_{yy}\right)\left[\omega_y\right][\omega_z] + L_x$$

$$I_{yy}\dot{\omega_y} = -(I_{xx} - I_{zz})[\omega_z][\omega_x] + L_y$$

$$I_{zz}\dot{\omega_z} = -\left(I_{yy} - I_{xx}\right)[\omega_x]\left[\omega_y\right] + L_z$$

Fig 1.1: Euler's Rotational EOM

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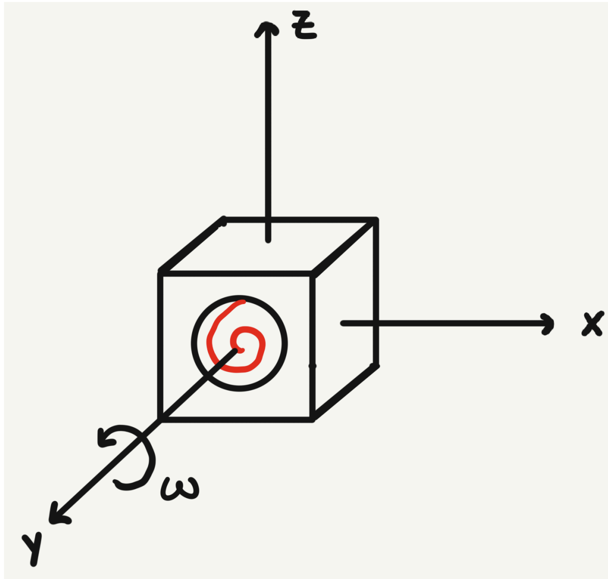

Fig 1.2: Physical Rate Gyro Coordinate Frame/Tilt Axes

PROBLEM 2 - CONTROL MOMENT GYROS (CMG)

a) **Applying Torque to Spinning Bicycle Wheel**
The wheel tries to resist the rotation of the axis and does it by spinning the person in the opposite direction of the rotation. So when the wheel is held out front the person and they try to apply a positive x-axis rotation (rotating the wheel clockwise w/ respect to the person holding the wheel). The counteracting moment would consequently rotate the person holding the wheel counterclockwise.

b) **Tracking a Target**
Assuming that the initial position was 0 degrees yaw:
- A target to the right (+yaw) required the user to tilt the wheel counterclockwise (-roll).
- A target to the left (-yaw) required the user to tilt the wheel clockwise (+roll).

c) **Inverting the Wheel**
Applying a positive roll moment about the x-axis (arms out in front) and trying to feel the resistive force while tilting along the y and z direction has force that causes it to resist and go further along. Applying the yaw moment and trying to rotate about the z-axis felt more resistive than rotating along the x and y axis. The rotation about the axes of the rotation of the wheel felt no change.

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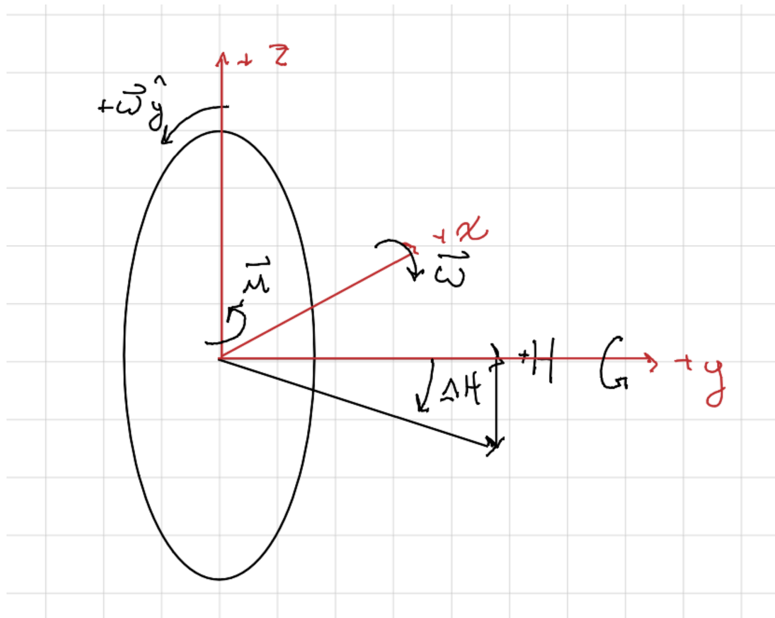

Fig 2.1: CMG with Momentum Vector, Rate of Change of Momentum and Resulting Moment Vector along given Axes

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PROBLEM 3: SPIN MODULE EXPERIMENTS

3.1) RATE GYRO MEASUREMENTS AND CALIBRATION

a)

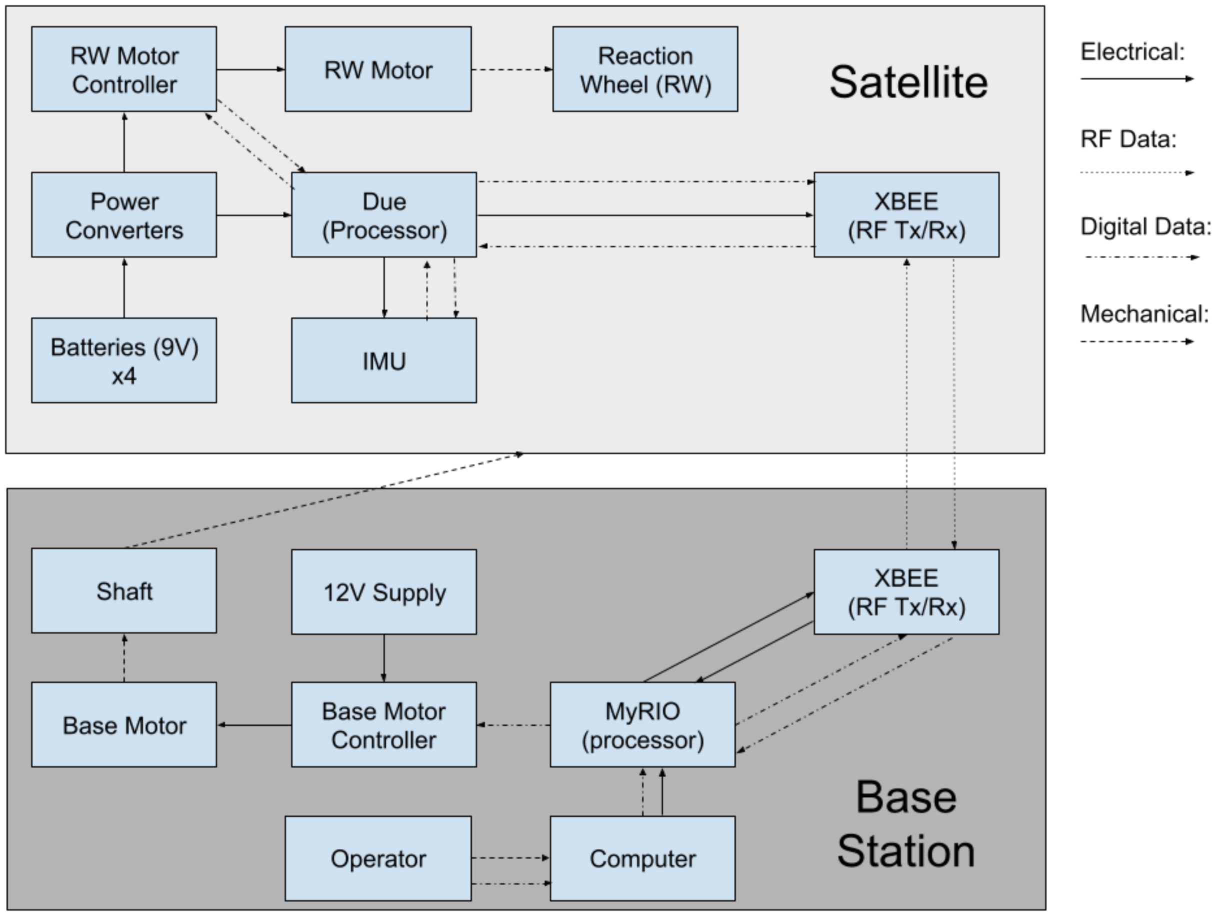

Fig 3.1: Functional block diagram representing key elements (Gyro, Power Supply, Up/Down Link Communications) and linkages between power/mechanical devices and communications.

RW Motor Controller: Sends inputs to and controls the RW Motor
RW Motor: Controls the reaction wheel.
Reaction Wheel (RW): Disk that instills an angular momentum on the satellite, allowing for the control of orientation and rotation in one dimension.
Power Converters: Processes and controls the voltage of the batteries to the correct voltages for each electrical component.
Due (Processor): Primary processor of the satellite unit.

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XBEE (RF Tx/Rx): Used for wireless communication between the satellite and base station.
Batteries (9V) x4: Batteries that power the satellite unit
IMU: Inertial Measurement Unit. Used to sense one axis of spin rotation rate/angle.
Shaft: Supports the satellite unit and allows for rotation of the satellite unit.
12V Supply: Powers the base unit.
Base Motor: Controls the shaft.
Base Motor Controller: Sends inputs to and controls the Base Motor.
MyRIO (processor): Primary processor of the base unit. Communicates via USB to the operator's computer running LabVIEW.
Operator: The user of the system. Determines what experiments will be performed and what data will be collected.
Computer: Displays the data collected through a GUI, can export the data for further processing and analysis.

b)

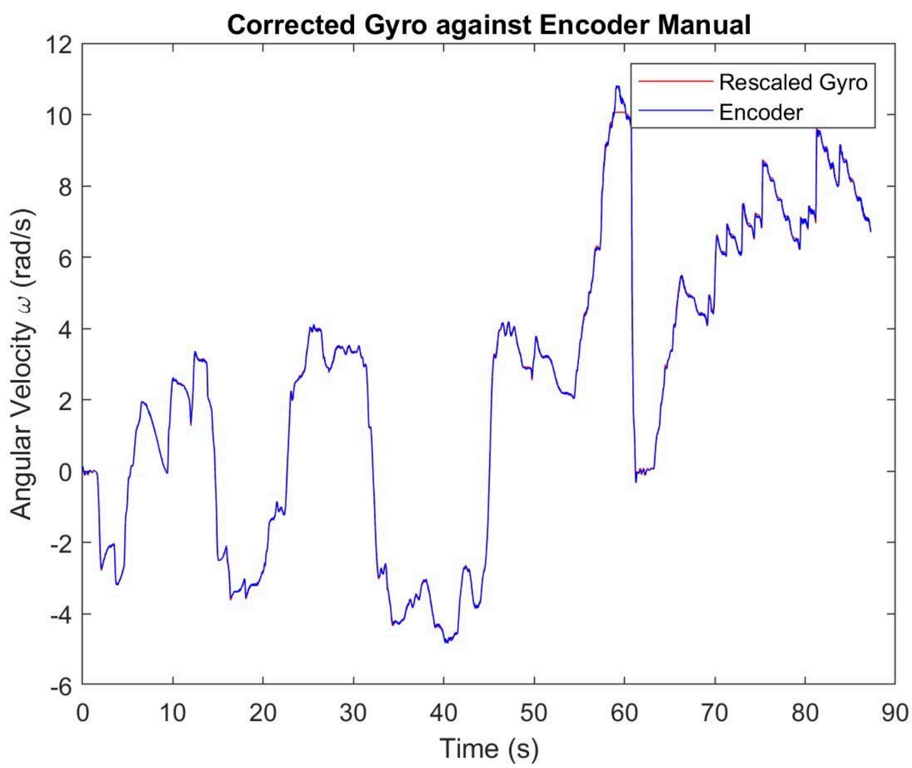

Fig 3.2: Manual Control Gyro Output Looping in relation to Base Encoder

The data recorded are taken every 10 ms time-interval during rate gyro measurements. While observing the recorded data for manual control, it shows that as the magnitude of input rate (in

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RPM) increases, gyro output (rad/s) also increases. The flip in direction of the gyro output is determined by its coefficient sign and with deep analysis of the recorded data we have observed that this usually takes place when input rate goes to zero and followed by the sign flip for gyro output that in general reflects the change in direction of rotation/angular velocity and vice-versa.

c)

Fig 3.3: Truth (Encoder Rate Measurement) and Measured (Gyro Rate Measurement) Relation at different time intervals

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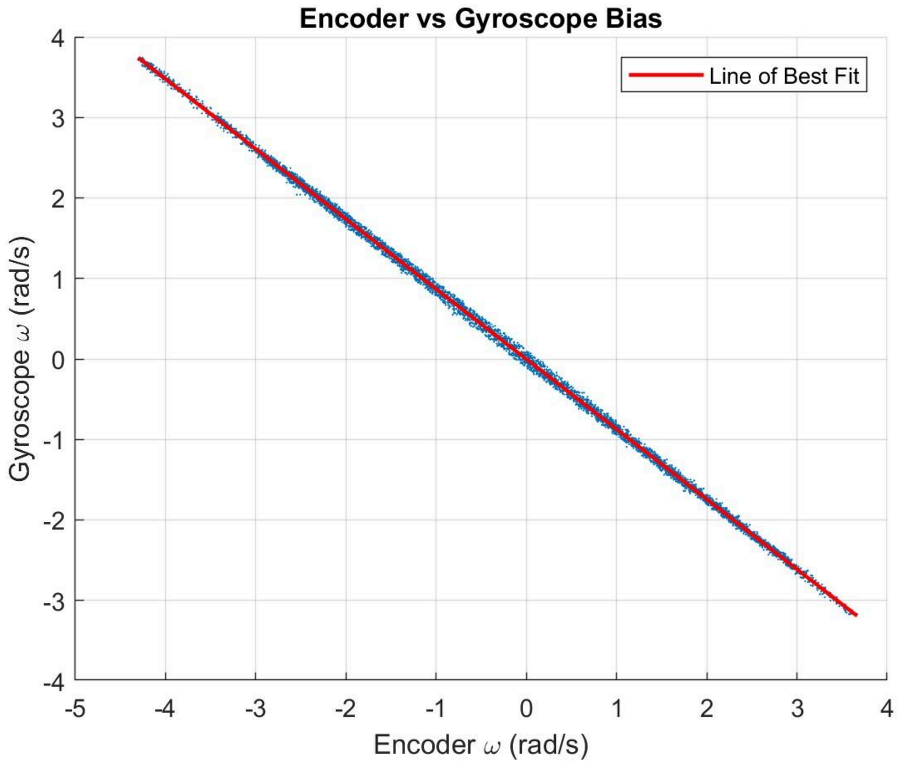

Fig 3.4: Linear Relationship for Angular Velocities between Truth (Encoder Rate Measurement) and Measured (Gyro Rate Measurement)

$$x' = \frac{(x-b)}{K} \quad \text{(Equation to correct gyro data)}$$

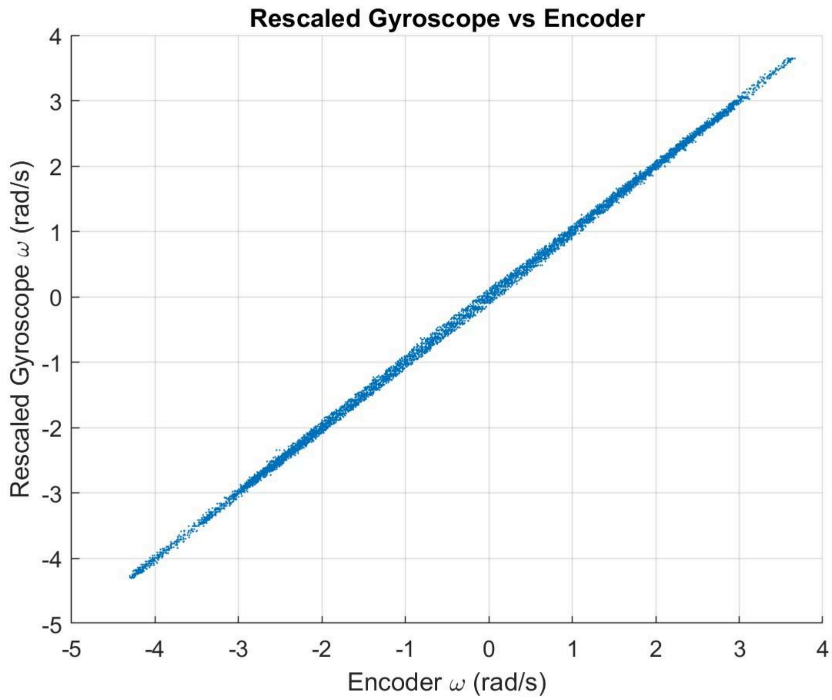

Fig 3.5: Calibrated Linear Relationship for Angular Velocities between Truth (Encoder Rate Measurement) and Measured (Gyro Rate Measurement)

Trials	Trial 1	Trial 2	Trial 3
Sensitivity (K)	0.8719	0.8709	0.8699
Bias (b) [rad/s]	0.0021	0.0014	0.0023

Fig 3.6: Bias and Sensitivity Measurement for multiple Gyro trials

Mean: 0.00193 rad/s Standard Deviation: 4.726×10^{-4} rad/s

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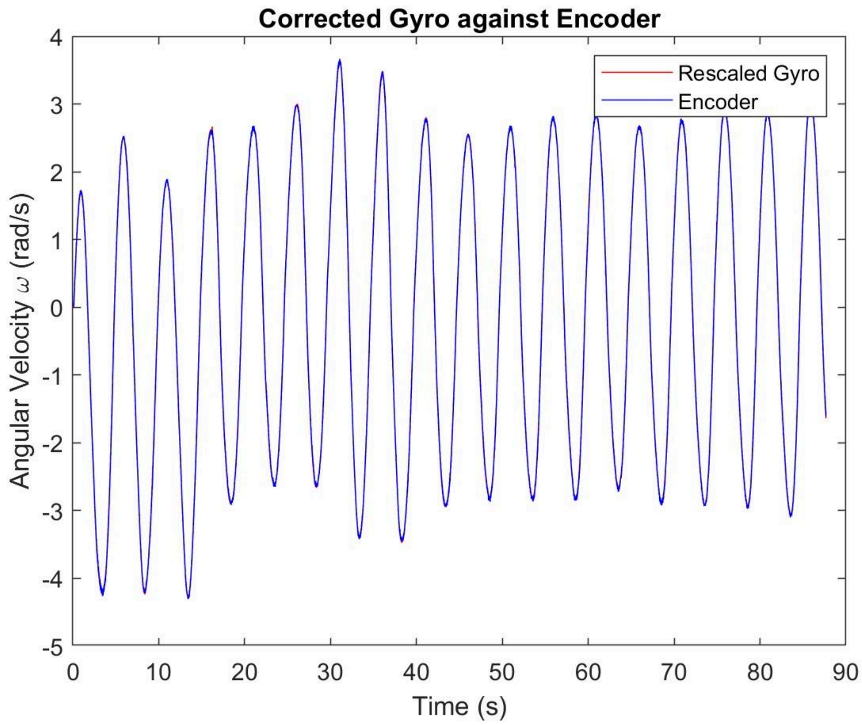

Fig 3.7: Time History Plot for the Encoder Rate measurements and Calibrated Rate

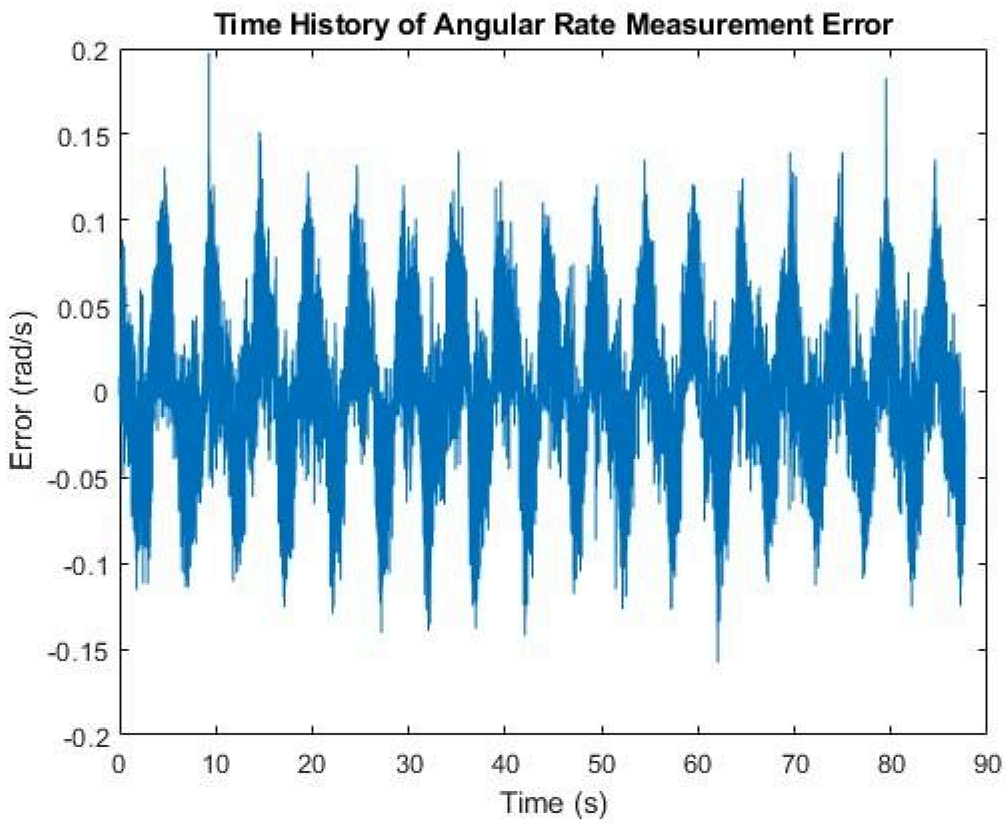

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Fig 3.8: Time History Plot of Angular Rate Measurement Error

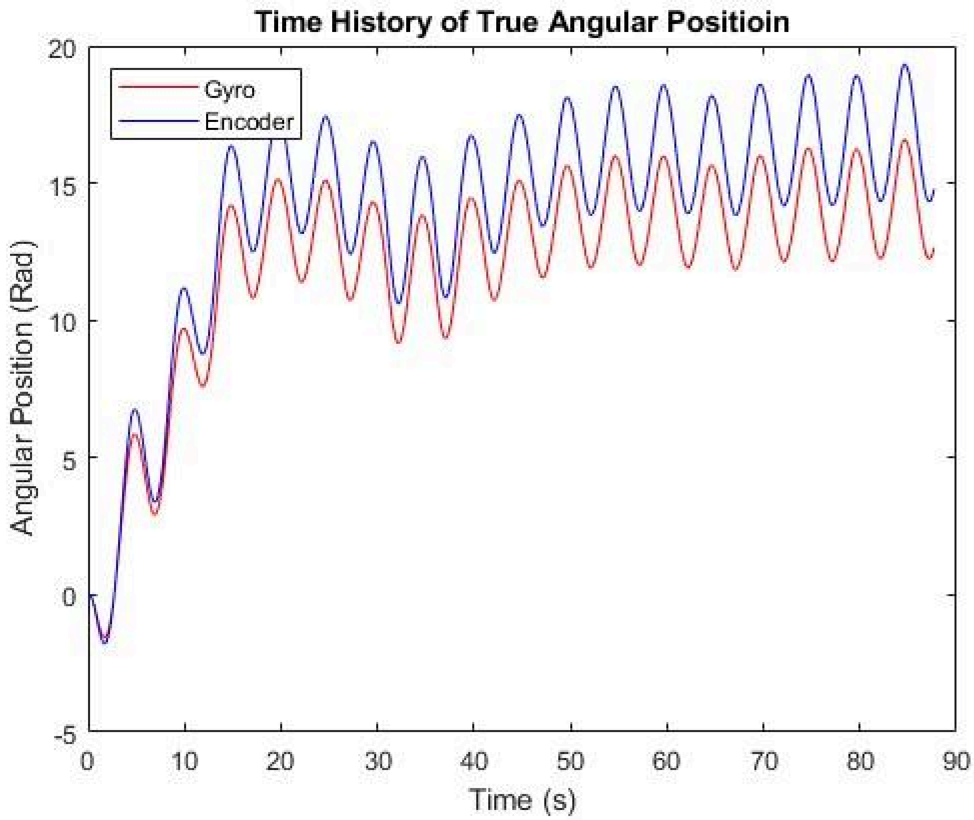

Fig 3.9: Time History Plot of True Angular Position w/ Both Values Positive to see Relationship

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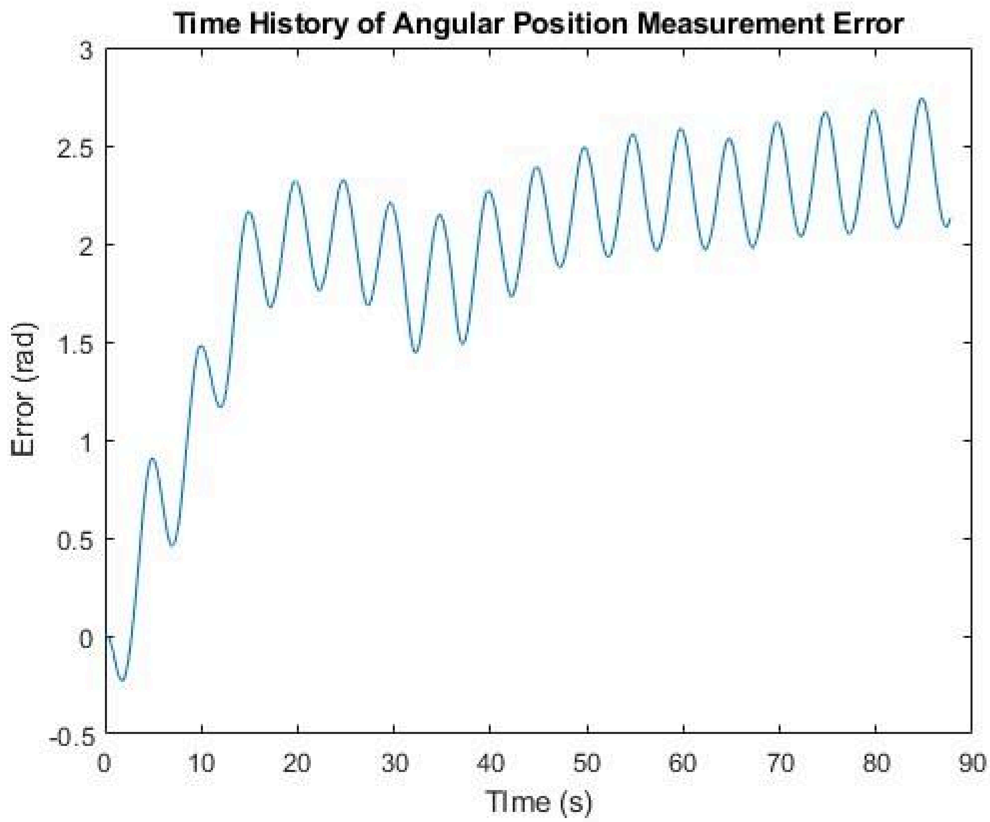

Fig 3.10: Time History Plot of Angular Position Measurement Error showing growing error

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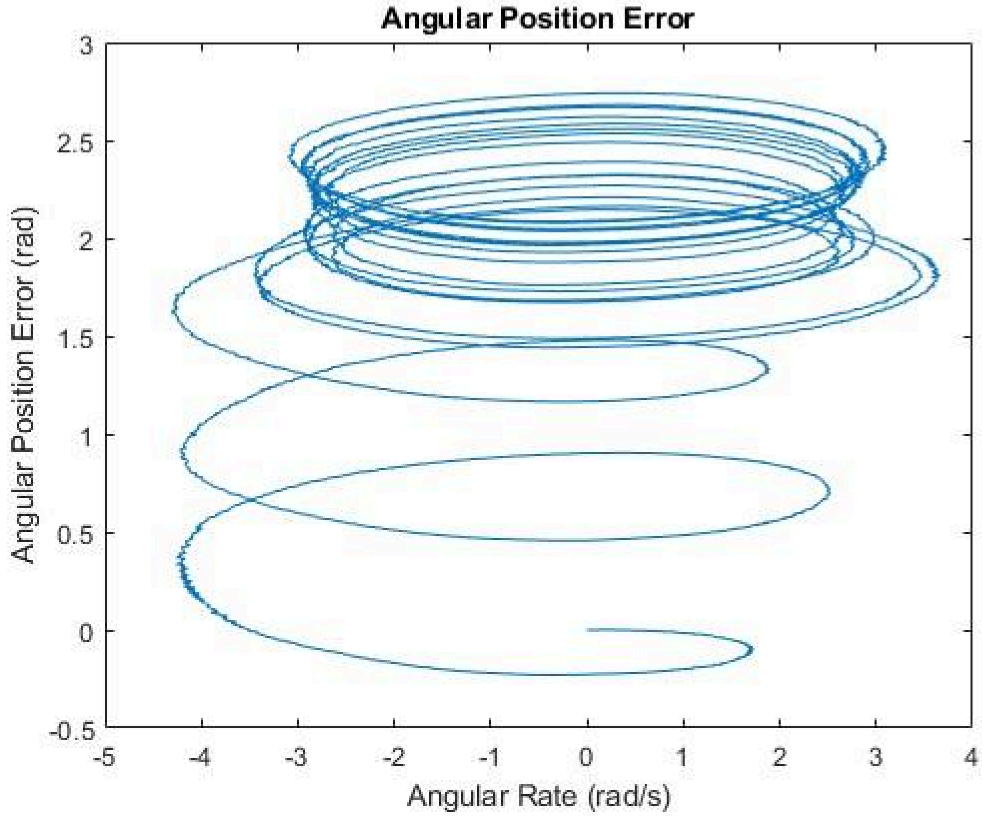

Fig 3.11: Angular Position Error Plot Against Angular Rate

d.)

The MEMS gyro was able to be measured to pretty decent sensitivity as it almost all showed pretty good linearity with very little outliers. Resulting in a very solid line of best fit as seen in figure 3.5. The bias was rather low compared to the objective measurements. Anywhere between -3 to 3 rad/s useful data since it has a good amount of scatter points and unlike past -3 rad/s where the data gets thin.

When given all the data files, the abrupt changes are easy to find since it results in non linear data, being most apparent at the tails of the data, appearing to curl and become non linear. The gyro bias is pretty easily repeatable and relatively the same value and doesn't change much between trials. This is an important measurement because it will give the data a slight skew towards a certain value. In this case it was all positive, so it was given a non-zero value for its initial value. So it needs to be corrected in order to get precise and correct measurement values.

The two values are misaligned by a certain factor after the bias is gotten rid of. This is where the values need to be rescaled by a factor to get the two sets to line up. This is when the rescale factor needs to be divided, which in this case is a linear value, thus meaning that the accuracy when going further away from the origin, the error will

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increase in a linear factor. It appears as if you have two gears that are recording the data and one is exactly 100 teeth while the other is close to 100 but not exactly. Resulting in one rotation is recorded as 2π, while the other rotation is recorded as slightly more or less than 2π. So consequently as more rotations are taken, more error per rotation is added. Thus the corrected gyro vs encoder looks like figure 3.7. Showing that the two when rescaled are the exact same signal. Figure 3.11 is interesting as it shows that initially the error starts off at zero but as the angular rate increases or decreases the rate at which error is accumulated increases. Meaning that as the angular rate increases, the rate at which the error accumulates also increases.

3.2) REACTION WHEEL MOI MEASUREMENT

a)

We used the LabVIEW VI to apply torques to our reaction wheel and collected data on time (ms), commanded torque (mNm), angular velocity (RPM), and actual current (A). While collecting this data we kept our aircraft fixed. For each trial, angular velocity was plotted against time and linear fit was applied to each plot to determine the angular acceleration for that specific torque. The torque and acceleration were then used to calculate the wheel's moment of inertia with I = torque / alpha. Four trials were completed for this section, yielding values of inertia I = 3.1607, 2.1546, 1.9976, 2.7035 (all $* \ 10^{-4}\ kgm^2$). From these, we get $I_{mean} = 2.5041 * 10^{-4} kgm^2$ and a standard deviation $I_{STD} = 5.3215 * 10^{-5} kgm^2$. This shows that there is a consistency among the four trials.

Using this mean inertia and standard deviation, the wheels ability to resist an aerodynamic torque of $1 * 10^{-4}\ Nm$ can be evaluated by using the angular acceleration, $\alpha = \tau/I$ and the time to reach. We know the maximum speed has a limit of 4000 rpm (which converts to about 418.9 rad/s). The time is equal to $t = I\omega/\tau = 17.5\ minutes$. The corresponding angular momentum capacity is found by solving $H = I\omega = 0.105\ Nms$.

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$\tau = 10^{-4}$ N·m

4000 rpm → 418.87902 rad/s

$I_{mean} = 2.5041 \cdot 10^{-4}$ kg·m²

$I_{std} = 5.3215 \cdot 10^{-5}$

$\alpha = \frac{\tau}{I}$ $\quad t = \frac{\omega_{limit}}{\alpha} \rightarrow \frac{I\omega_{limit}}{\tau} = \frac{(2.5041 \cdot 10^{-4})(41887902)}{10^{-4}} = 1{,}048 \text{ s} \rightarrow 17.48 \text{ mins}$

$H = I\omega_{limit} = (2.5041 \cdot 10^{-4})(418.87902) = 0.1049 \text{ N·m·s}$

$\sigma_t = \frac{\omega_{limit}}{\tau} I_{std} \rightarrow 222.9 \text{ s} \rightarrow 3.71 \text{ min}$

$\sigma_H = \omega_{limit} I_{std} \rightarrow 0.022 \text{ N·m·s}$

time is 17.5 mins (± 3.7 mins)

angular momentum = 0.1049 NMS (± 0.022 N·m·s)

Fig 3.12: Workings for the (i) time [minutes] (ii) angular momentum of the wheel [Nms]

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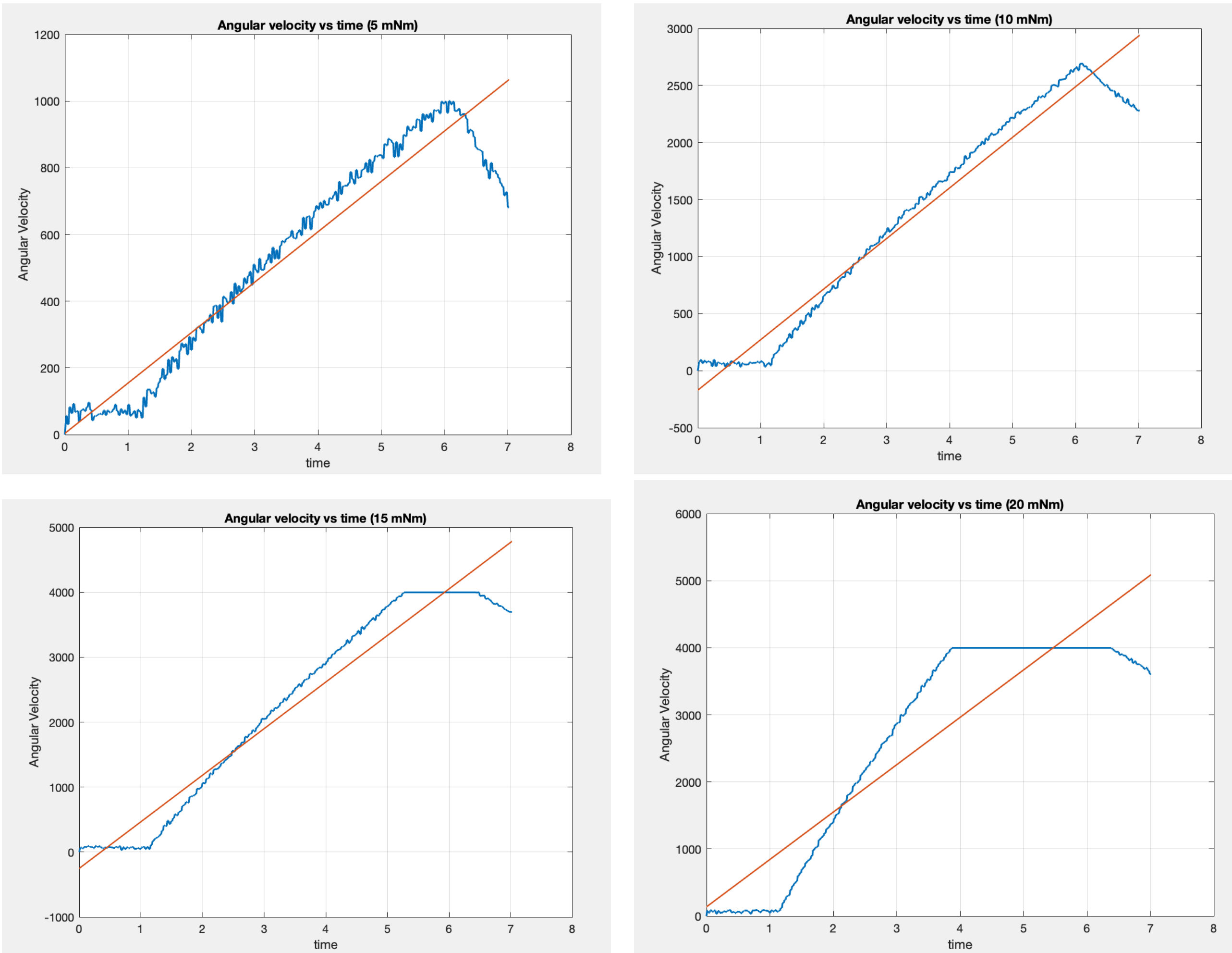

Fig 3.13: MOI Calculations with applying constant torque [5, 10, 15, 20] mNm throughout four different experiments and recording Angular Velocities data as wheel accelerates

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3.3) SPACECRAFT CONTROL

a) Spacecraft Moment of Inertia

Average Torque(τ)	Angular Acceleration(α)	Spacecraft MOI (I)
3.0mNm	**-0.2136 rad/s^2**	**0.0139 kgm^2**
6.0mNm	**-0.463 rad/s^2**	**0.0129 kgm^2**
9.1mNm	**-0.936 rad/s^2**	**0.0097 kgm^2**

Average Moment of Inertia = 0.0122 kgm^2

Using the spacecraft mechanism, data was recorded and then processed using MATLAB to get the values in the table above.

b) Explore Closed-Loop PD Gains

i)

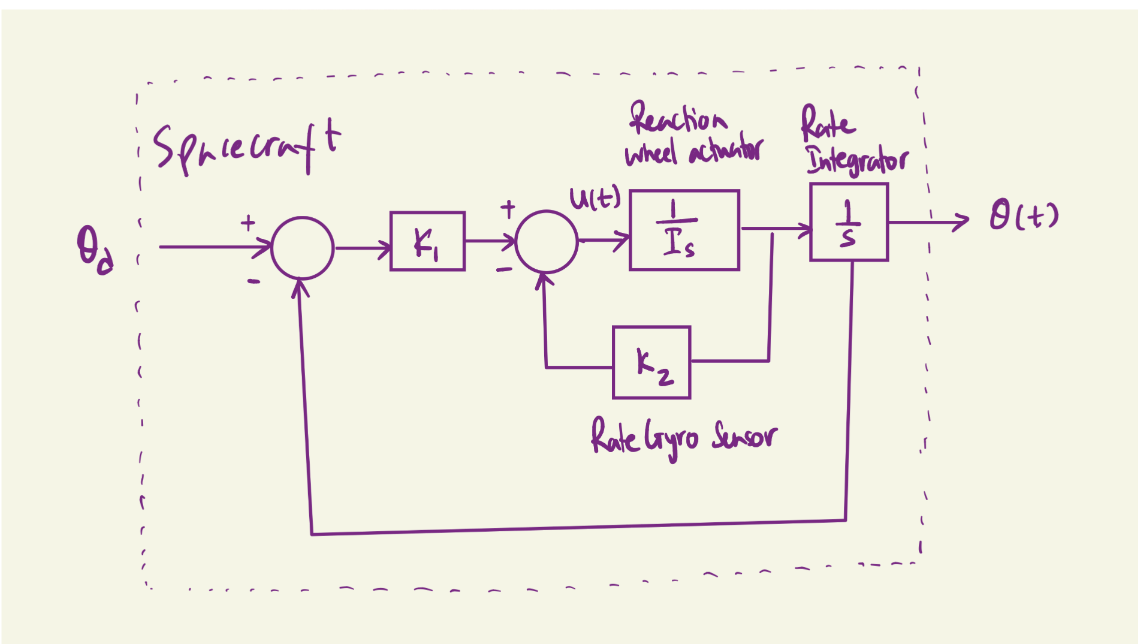

Fig 3.14: Block Diagram (1 DOF) comprising a Spacecraft, Reaction Wheel Actuator, Rate Gyro Sensor, Rate Integrator, PD Control Law (w/ K1 and K2 Gain)

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ii)

$$\alpha = I\ddot{\theta} = u(t)$$

$$u(t) = -K_p(\theta - \theta_d) - K_d\dot{\theta} \qquad \text{Note: want } \theta(t) = \theta_d \text{ , } \dot{\theta} = 0$$

$$I\ddot{\theta} = -K_p(\theta - \theta_d) - K_d\dot{\theta} \qquad \text{where , } K_1 = K_p \text{ ; } K_2 = K_d$$

$$\boxed{\ddot{\theta} = -\frac{K_1}{I}(\theta - \theta_d) - \frac{K_2}{I}\dot{\theta}}$$

Fig 3.15: Dynamical System Equation that describes the spacecraft rotating about single spin axis (One-Dimensional)

iii)

K1 Value	**Description of Motion**
27	Slow response to the rotating gyro with nearly 1 complete rotation.
70	Slightly Quicker response to the rotating gyro but seems unnoticeable.
200	Quicker response to the rotating gyro align with sinusoid motion

Fig 3.16: Observations with changing Proportional Gain [0 200] and describing the motion of the Control System (assume S/C)

iv) We can choose the gain values, by just requiring that it needs a proportional value and nothing else. This would give a system that feeds in a K1 gain and no K2 gain as we aren't trying to correct the attitude to a certain reference value. So a higher K1 would result in a stronger response to slowing down the system. This might not be the best as it would likely overshoot, so figuring out a K1 gain that allows a decent exponential decay without too much oscillation would be ideal.

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v)

K1 Value	**K2 Value**	**Description of Motion**
100	15	Nominal oscillations rate and few “to and fro” deviating motion about the mean position; lesser time to settle.
200	15	Quicker oscillations and taking longer to settle.
100	40	Overdamped (no oscillations
100	5	Many oscillations, but marginal improvement in settling

Fig 3.17: Observations with changing K1 [0 200] and K2 [0 40] and describing the motion of the Closed-Loop Feedback System in terms of damping effect.

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c) Design Feedback Gains

Maximum Overshoot Equation: $M = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}}$

10% max overshoot $\Rightarrow M = 0.1$

Derivation of ζ:

$$\ln(0.1) = \frac{-\pi\zeta}{\sqrt{1-\zeta^2}}$$

$$\ln(0.1)\sqrt{1-\zeta^2} = -\pi\zeta$$

Square both sides:

$$\ln(0.1)^2(1-\zeta^2) = \pi^2\zeta^2$$

$$\ln(0.1)^2 - \zeta^2\ln(0.1)^2 = \pi^2\zeta^2$$

$$\ln(0.1)^2 = \zeta^2(\pi^2 + \ln(0.1)^2)$$

$$\zeta^2 = \frac{\ln(0.1)^2}{\pi^2 + \ln(0.1)^2}$$

$$\zeta = \sqrt{\frac{\ln(0.1)^2}{\pi^2 + \ln(0.1)^2}} = 0.591$$

Settling Time Equation: $t_s = \frac{-\ln(p)}{\zeta\omega_n}$

5% settling time $\Rightarrow p = 0.05$, $t_s = 1.5\,s$

Derivation of ω_n:

$$1.5 = \frac{-\ln(0.05)}{\zeta\omega_n}$$

$$\omega_n = \frac{-\ln(0.05)}{\zeta \cdot 1.5} = \frac{-\ln(0.05)}{0.591 \cdot 1.5} = 3.378\,\frac{rad}{s}$$

Equation of Motion: $\ddot{\theta} + \frac{k_2}{I}\dot{\theta} + \frac{k_1}{I}\theta = 0 = \ddot{\theta} + 2\zeta\omega_n\dot{\theta} + \omega_n^2\theta$

$$\frac{k_2}{I} = 2\zeta\omega_n \Rightarrow k_2 = 2 \cdot 0.591 \cdot 3.378 \cdot 1.278 = 5.104$$

$$\frac{k_1}{I} = \omega_n^2 \Rightarrow k_1 = 3.378^2 \cdot 1.278 = 14.59$$

Fig 3.18: Closed Loop System with 5% settling time and Overshoot less than 10%

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From this derivation, we see that K1 = 14.59 Nm/rad and K2 = 5.104 Nm/(rad*s)

d) Test Feedback Gains

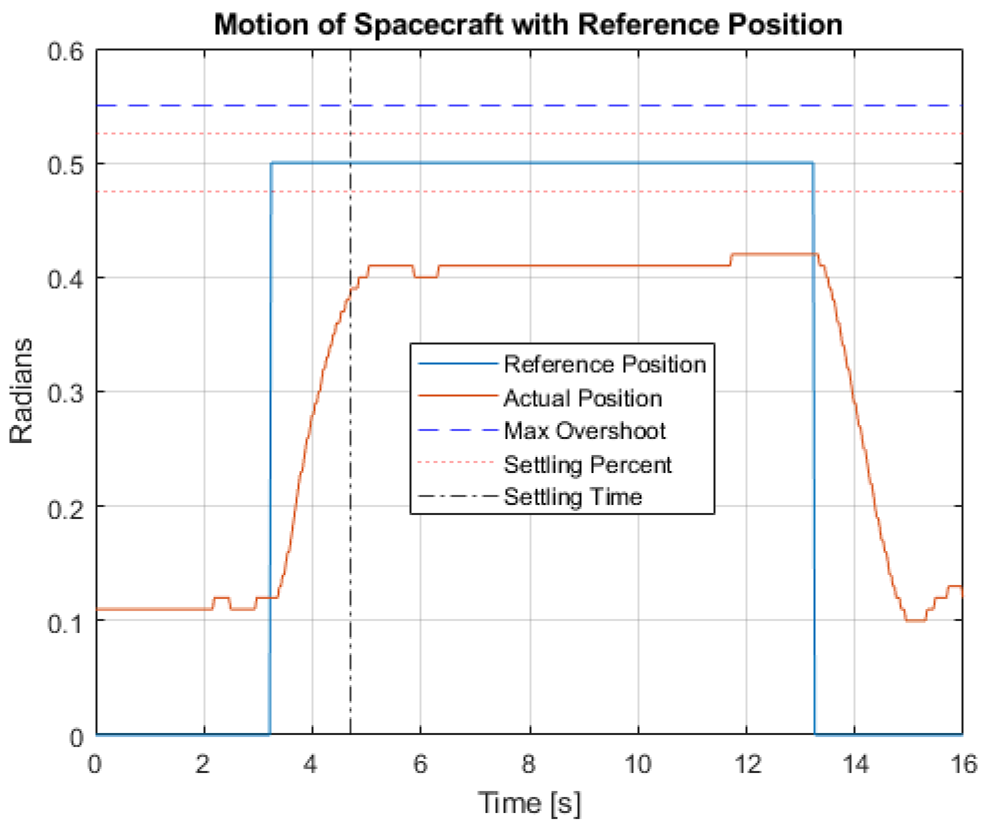

Fig 3.19: Physical Test of Derived K-Values

When the K-values from 3.3c were utilized in the physical system, we see that it successfully stayed within overshoot bounds but failed to reach the settling percentage at all, let alone within the allotted settling time. This is likely due to the fact that the derived K-values did not account for the physical friction and resistance to motion within the system, causing the actual damping ratio to be greater than calculated. This can be seen with the high settling value error.

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APPENDIX: MATLAB CODE

3.1c code:

```
clc; clear all; close all;

raw_data = readmatrix("2025_09_23_003_AUM3");
raw_data = raw_data((2:length(raw_data)), :);       % Delete the first row of
useless data
% Declare all the different variables
time = raw_data(:, 1);          % s (Data is kinda messed up cause the time is
way wack)
% Conditioning the time so that it all starts @ 0s and graphing is nice
time = time-time(1);            % s
gyro = raw_data(:, 2);          % Rad/s
encoder_RPM = raw_data(:,3);    % RPM
% Convert the RPM to rad/s
encoder_Rads = (encoder_RPM./60).*(2*pi);  % Converted into Rad/s

% Plotting the time history of the encoder and the gyro
figure();
plot(time, gyro, "Color", "Red")
hold on;
plot(time, encoder_Rads, "Color", "Blue")
grid on;
legend("Gyroscope", "Encoder")
xlabel ("Time (s)")
ylabel ("Angular Velocity \omega (rad/s)")
title("Time History of Encoder and Gyro")


% Plotting the Encoder vs the Gyro rate to calibrate
figure();
scatter(encoder_Rads, gyro, 0.2)    % Tail has a sort of non-linear fashion
about it
hold on;


% Taking a line of best fit for the values on the plot above
poly_vals = polyfit(encoder_Rads, gyro, 1);
LoBF = polyval(poly_vals, encoder_Rads);


% Plotting the line of best fit on the graph
plot(encoder_Rads, LoBF, "Color", "Red", LineWidth=1.5);
grid on;
xlabel("Encoder \omega (rad/s)")
ylabel("Gyroscope \Omega (rad/s)")
```

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```
title("Encoder vs Gyroscope Bias")
legend("", "Line of Best Fit")


% Rescaling the Gyroscope Data
rescale_gyro = ((gyro - poly_vals(2))./poly_vals(1));


figure();
scatter(encoder_Rads, rescale_gyro, 0.2)
grid on;
xlabel("Encoder \omega (rad/s)")
ylabel("Rescaled Gyroscope \Omega' (rad/s)")
title("Rescaled Gyroscope vs Encoder")
figure();
plot(time, rescale_gyro, "Color", "red")
hold on;
plot(time, encoder_Rads, "Color", "blue")
legend("Rescaled Gyro", "Encoder")
xlabel("Time (s)")
ylabel("Angular Velocity \omega (rad/s)")
title("Corrected Gyro against Encoder")

% Standard Deviation and Mean calculations
bias = [0.0021, 0.0014, 0.0023];
standard_dev = std(bias);

% Plotting the rate measurement error
error_1 = rescale_gyro - encoder_Rads;
figure();
plot(time, error_1)
xlabel("Time (s)")
ylabel("Error (rad/s)")
title("Time History of Angular Rate Measurement Error")


%% Angular Position
% Integrating Rad/s to get Rad measurement
theta_gyro = cumtrapz(time, gyro);
theta_gyro_unwrapped = unwrap(theta_gyro);

% Unwrapping so that instead of looping back it will show growing error
theta_encoder = cumtrapz(time, encoder_Rads);
theta_encoder_unwrapped = unwrap(theta_encoder);

% Plotting Angular Position vs Time
figure();
```

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```
plot(time,theta_gyro_unwrapped, "Color", "Red")
hold on;
plot(time,-theta_encoder_unwrapped, "Color", "Blue")
xlabel("Time (s)")
ylabel("Angular Position (Rad)")
title("Time History of True Angular Positioin")
legend("Gyro", "Encoder", "Location", "northwest")

% Plotting Angular Position Error
error_2 = (-theta_gyro_unwrapped) - (theta_encoder_unwrapped);
figure();
plot(time, error_2)
xlabel("TIme (s)")
ylabel("Error (rad)")
title("Time History of Angular Position Measurement Error")

% Plotting Angular Position as a function of Encoder Rate Measurement
figure()
plot(encoder_Rads, error_2)
xlabel("Angular Rate (rad/s)")
ylabel("Angular Position Error (rad)")
title("Angular Position Error")
```

3.2a code:

```
clc;
clear;
close all;
% data files and torques
files = {
   '2025_09_23_003_T5t5'
   '2025_09_23_003_T10t5'
   '2025_09_23_003_T15t5'
   '2025_09_23_003_T20t5'
};
labels= {'5 mNm','10 mNm','15 mNm','20 mNm'};
% mNm to Nm
torques = [5 10 15 20] * 1e-3;
alphas = zeros(length(files), 1);
% import the data and plot angular velocity vs. time
for i = 1:length(files)
   %read in to the loop
   data = readmatrix(files{i});
   t = data(:,1)*1e-3; % getting this in seconds
   omega = data(:,3); % getting this in RPM
   % fit a line to estimate angular acceleration
   fitline = polyfit(t, omega, 1);
```

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```
    alpha_rpms = fitline(1);
    alphas(i) = alpha_rpms * (2 * pi / 60); % rad/s^2
    % plot data
    figure;
    plot(t, omega, 'LineWidth',1.5);
    hold on;
    plot(t, polyval(fitline,t), 'LineWidth',1.5);
    xlabel('time');
    ylabel('Angular Velocity');
    title(['Angular velocity vs time (' labels{i} ')']);
    grid on;
    % compute inertia estimate for each trial
    I_trials(i) = torques(i) / alphas(i);
end
% mean and standard deviation of inertia
% inertia from slope of torque vs alpha
p = polyfit(alphas, torques, 1);
I = p(1);
% I_trials is the 4 separate inertia estimates so...
I_mean = mean(I_trials);
I_std = std(I_trials);
```

3.3c code

```
%% Getting moment of inertia
moment_data = readmatrix("2025-10-07-003-t6_33a");
%plot(moment_data(3:length(moment_data), 1), moment_data(3:length(moment_data),
3))

data_chop = moment_data(102:252, :);
data_chop(:,1) = data_chop(:,1)./1000;      % Put time into seconds
data_chop(:,3) = ((data_chop(:,3).*2.*pi)./60);     % Put the RPM to rad/s

plot(data_chop(:,1), data_chop(:,3))                 % Plotting rad/s to time
torque = data_chop(:,4).*25.5;                       % Converting the amps into
a readable torque value
avg_torque = mean(torque)*(10^-3);                   % Grabbing the average
torque then converting into the mNm

vals = polyfit(data_chop(:,1), data_chop(:,3), 1);       % Fitting the data
into the downslope, kind of need to hardcode the slope and data points

moment_inertia = avg_torque/abs(vals(1))    ;  % I = L/a for the torque
calculations
```

3.3d code

```
clc; clear; close all
% load in data
```

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```
data = readmatrix("2025_003_3d_updated");
% clean up data
data = data(3600:5200,:); %choose important part of the run
time = data(:,1);
% set time = 0 for run, turn it from ms to s
time = (time-time(1))/1000;
ref = data(:,2);
pos = data (:,3);
% creat graph of data
figure();
plot(time, ref)
hold on;
grid on;
plot(time, pos)
title("Motion of Spacecraft with Reference Position")
ylabel("Radians")
xlabel("Time [s]")
yline(0.55, "--b")
yline(0.525, ":r")
yline(0.475, ":r")
xline(4.71, "-.k")
legend("Reference Position","Actual Position","Max Overshoot","Settling
Percent","","Settling Time")
```

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Team Member Participation Table
Lab # 3801 Lab 3-11
Date Oct 12, 2025

	Plan	Model	Experiment	Results	Report	Code	ACK
Name							
Hailey Ball	2	2	1	2	1	2	X
Gavin Bull	1	2	2	2	1	1	X
Jayden Van Dam	1	1	1	2	2	2	X
Ritik Sarraf	2	1	2	2	2	1	X

For each team member enter: 2 = Lead, 1 = Participate, 0 = Not involved, for each element.
If an element is not relevant for a particular lab assignment, just put N/A in that column.
You are welcome to provide additional comments about team member participation below the table.

Lab Participation Elements

Plan – Planning and Discussion
Discuss the overall assignment, identify key tasks and lead member for each task, develop a plan for completion of the assignments. In each lab period, assess progress, review/correct work, and adjust planning as necessary.

Model – Theory and Modeling
Draw diagrams, derive models/equations, compute approximate/expected results (used to check experimental or simulation results), including preparation of relevant sections

Experiment
Draw diagrams, describe procedure, identify measurements and expected errors/uncertainties, conduct the experiment and note conditions or issues

Results and Analysis
Plot results, compare to theory, compute and tabulate statistics, analyze errors in terms of magnitude and sources

Report
Prepare overall lab report according to requirements; gather inputs for each section including participation table; write abstract and acknowledgments. Edit for technical accuracy, clarity, language (grammar and spelling), and format.

Code
Write, comment, test code used to produce the results and plots in the report.

ACK
An X in this column means the listed student ACKNOWLEDGES that they participated in the creation of the table and have seen the final version before it was submitted. Placing an X for a student who has not seen the table is consider a violation of the Honor Code and will result in a 0

for the assignment for the team members who completed the table without their lab partners concurrence. If a student did not see the final table, leave this column blank.

Participate means that you did some of the work for an element, including writing up the relevant portion of the report. *Lead* means that you took primary responsibility for this element; it does not mean that you did the entire element yourself.

We expect each student to participate in multiple elements of the assignment and to take the lead on one or two elements of each assignment. To get the most out of the course, students should strive to vary their leadership and participation over the course of the semester.